

Programming Project / C++

Table Tennis Tournament Management System

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I. Task Description

Application 1 is responsible for managing players and matches:

- Add matches
- Modify match scores
- Delete matches
- View rankings
- View all matches

Application 2 is responsible for player interaction:

- View personal match history
- Search players
- View rankings
- Filter matches by player name

Both applications communicate through shared text files.

II. Data Structures Used

The following classes will be used:

```
class Match {
    string player1;
    string player2;
    int score1;
    int score2;
};

class Player {
    string name;
    int wins;
    int losses;
```

```
    int totalPoints;  
};
```

III. File Structure

matches.txt

Stores all played matches.

Format:

```
<player1> <score1> <score2> <player2>
```

Example:

```
Ali 21 17 Reza  
Sara 21 19 Mina
```

ranking.txt

Stores ranking information.

Format:

```
<player_name> <wins> <losses> <points>
```

Example:

```
Ali 3 1 63  
Sara 2 0 42
```

IV. Interacting with Executables

Application 1 Commands

```
./app_1.exe add_match <player1> <score1> <score2> <player2>
```

Adds a new match.

```
./app_1.exe delete_match <player1>
```

Deletes a match.

```
./app_1.exe view_matches
```

Displays all matches.

```
./app_1.exe generate_ranking
```

Generates player rankings.

Application 2 Commands

```
./app_2.exe view_rankings
```

Displays rankings.

```
./app_2.exe search_player <name>
```

Searches for a player.

```
./app_2.exe player_history <name>
```

Displays player's match history.

Important Note

The project will use:

- File handling
- Classes
- Functions
- Command line arguments
- Data persistence between runs